Templated Problem Demo Algebra 2 / Calculus Version A

Name: ____

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1. (12 pts) Let $f(x) = (x+3)(x-1)(x-4)$.

a. Find all real zeros of f.

b. Determine the following key features:

i. The y-intercept. **ii.** Where $f(x) > 0$.

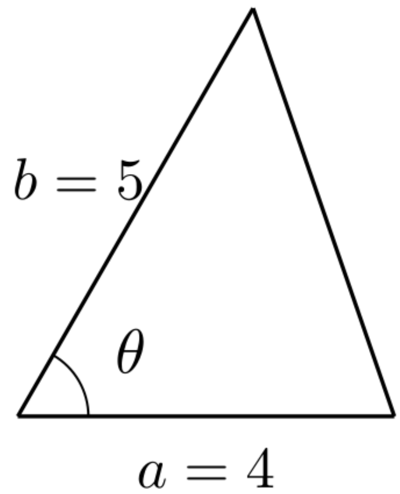
c. Describe the end behavior of f:

i. As $x \to -\infty$:

ii. As $x \to +\infty$:

2. (10 pts) The area of a triangle with sides a and b and included angle θ is $A = \frac{1}{2}ab\sin(\theta)$.

1. If $a = 2$, $b = 3$, and θ is increasing at $\frac{d\theta}{dt} = 0.2$ rad/s, find a formula for $\frac{dA}{dt}$ in terms of θ.



2. Find the rate at which the area is changing when $\theta = \pi/3$.